

Needs Analysis

Overdrive

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Summary

OverDrive – Needs Analysis

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2- Subscription needs (analysis & benchmark)

- 1- Identified services (F1 TV, Rally TV, WEC TV, MotoGP)
- 2- Estimate of the monthly cost of subscriptions

3- Hardware / physical equipment requirements

- 1- VR Headset (Meta Quest 3S)
- 2- Development computers
- 3- Estimated total material cost

4- Cloud infrastructure needs

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- 2- MVP option: simple VPS (OVH / Hetzner)
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5. Tools & services

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Introduction

This document presents the analysis of the needs necessary for the development and operation of OverDrive. The objective is to clearly identify the resources essential to the project, whether technical, material or organizational, while distinguishing the needs of a student MVP and those of a long-term scalable version.

OverDrive being a VR/AR and real-time oriented product, its needs are mainly divided between :

- Benchmark subscriptions (competitive analysis),
- development equipment (VR + workstations),
- a cloud infrastructure for the backend,
- coordination tools and services.

1- Infrastructure, service and equipment needs

Development and operation of OverDrive requires multiple resource categories :

- **Subscriptions to streaming platforms**, used solely to analyze existing user experience and compare against market standards.
- **Cloud infrastructure**, essential for hosting the backend, database, and real-time system.
- **Technical services**, including hosting, monitoring, and development tools.
- **Physical equipment**, including a VR headset and computers capable of supporting Unity and real-time development.

This analysis makes it possible to frame the costs of the project and to validate the feasibility in an MVP framework.

2- Subscription needs (analysis & benchmark)

In order to test existing platforms and understand the current standards of motorsport, several subscriptions are necessary. Their objective is strictly limited to :

- Analysis of the official experience,
- benchmarking of features (multi-camera, live timing, replays),
- comparison of broadcast formats according to disciplines.

These subscriptions are not used for direct integration into OverDrive, but for competitive study and UX validation.

1- Identified services

F1 TV

- Access to live timing data,
- replays,
- multi-camera,
- leading F1 market reference.

Rally TV

- Access to rally data,
- comparison of formats and distribution methods.

FIA WEC TV

- Objective: Endurance and multi-category benchmarking,
- useful for testing more complex formats.

MotoGP VideoPass

- Data + multi-angle,
- considered a UX benchmark in the motorsport/motorcycle world.

2- Monthly subscription cost estimate

Service	Estimated cost
F1 TV	6 €
Rally TV	10 €
FIA WEC TV	32 €
Moto VideoPass	?€

Estimated total: €48 to €60 / month.

A reduction in this cost is possible if certain subscriptions are used only over short periods (e.g. one month of benchmark), or if MotoGP is discussed according to priorities.

3- Hardware / physical equipment requirements

Material needs represent the main initial expense of the project, because OverDrive involves VR development, more demanding than a conventional mobile or web project.

1- VR headset

The recommended headset for the V1 is :

Meta Quest 3S

Estimated price: ~**300€**

This choice is motivated by several factors:

- Compatibility with Unity,
- excellent performance/price ratio,
- large user base,
- standalone headset (no need for a dedicated VR PC).

As part of an MVP, a single helmet is considered the minimum viable to test the immersive experience and iterate quickly.

2- Development computers

The project requires positions capable of :

- Develop under Unity,
- Manage backend tools,
- Run VR builds,
- Test 3D and real-time environments.

Hypothesis retained :

- 3 workstations

Estimated budget :

- €1200 per machine
- Total: $3 \times €1200 = €3600$

Recommended configuration :

- High-performance CPU (Ryzen 7 / i7)
- 16 GB RAM minimum
- Dedicated GPU (RTX 3060 or equivalent)
- NVMe SSD

3- Estimated total material cost

- VR headset: ~€300
- 3 computers: €3600

Estimated total equipment : ~**3900€**

4- Cloud infrastructure needs

OverDrive requires a server infrastructure to support :

- The API backend (REST),
- the real-time WebSocket server,
- the PostgreSQL database,
- the Redis cache,
- object storage (assets, optional replays).

1- MVP option: Simple VPS (OVH / Hetzner)

For a student MVP, a simple and economical solution is sufficient.

Estimate:

- High-performance VPS: €15–30/month
- Backend + database on the same machine
- Local Redis

Estimated total: **20–40€/month**

Advantages:

- Low cost,
- simple deployment,
- accessible maintenance.

This approach is adapted for a first version, with a reduced team and a limited number of users.

5- Tools & services

Several tools necessary for the operation of the team can be used free of charge in an MVP framework :

- **GitHub (free for standard use):** version control, issues, projects, pull requests.
- **Discord (free):** daily communication.
- **Google Meet (free):** weekly meetings.

Ces outils suffisent pour une organisation agile et un pilotage propre du projet.

6- Overall cost estimate

1- Investissement initial

Poste	Estimated cost
VR headset	300 €
3 computers	3600 €

Total initial estimé : **~3900€**

2- Coût mensuel récurrent (MVP)

Poste	Estimated cost
Benchmark subscriptions	50 €
Cloud infrastructure	80 €

Estimated monthly total: **~80€ / month**

A variation is possible according to :

- the actual use of subscriptions,
- the choice of VPS,
- the addition or not of additional services.

7- Analyse stratégique

1- Recommended approach for a student project / MVP

For an MVP, the goal is to limit costs while ensuring :

- a stable development environment,
- real VR testing capabilities,
- sufficient infrastructure for a functional prototype.

Recommendations :

- A **simple VPS** is more than sufficient,
- a VR headset is essential,
- subscriptions are only for benchmarking and can be activated temporarily.

In this context, costs remain under control.

2- Projection for a scalable version

If OverDrive evolves to a more ambitious version, several elements will have to be reinforced :

- Managed cloud infrastructure,
- advanced monitoring,
- enhanced security,
- separation of services (API, DB, Redis, storage).

In this case, the monthly budget could increase, but remain proportional to product growth.

Conclusion

Needs analysis shows that OverDrive's main expenses are related to VR development, in particular :

- the acquisition of a headset,
- the provision of workstations adapted to Unity and real-time processing.

The cloud infrastructure remains moderate for an MVP, with a simple VPS option sufficient. Subscriptions to motorsport platforms are useful for achieving a precise competitive benchmark, but they are not necessary continuously.

Overall, the initial investment remains reasonable for an immersive technology project, and the monthly costs are compatible with a prototype or student development phase.